

1. Administrative & Study Identification

Full Study Title: Study of Pembrolizumab (MK-3475) or Placebo Given With Best Supportive Care in Asian Participants With Previously Treated Advanced Hepatocellular Carcinoma (MK-3475-394/KEYNOTE-394) **Official Title:** A Phase III Randomized Double-blind Study of Pembrolizumab plus Best Supportive Care vs. Placebo plus Best Supportive Care as Second-Line Therapy in Asian Subjects with Previously Systemically Treated Advanced Hepatocellular Carcinoma (KEYNOTE-394) **Short Title/Acronym:** KEYNOTE-394 **Posting ID:** 541362913489557453 **Protocol Number:** MK-3475-394 **NCT Number:** NCT03062358 **Sponsor / Company:** Merck **Trial Status:** COMPLETED **Phase of Development:** PHASE3 **Investigational Product:** Pembrolizumab (MK-3475 / KEYTRUDA®) **ATC Code:** L01FF02 **EMA URL:** [Keytruda EPAR Product Information](#)

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To determine the efficacy and safety of pembrolizumab or placebo given with best supportive care (BSC) in Asian participants with previously systemically treated advanced hepatocellular carcinoma (HCC). The primary hypothesis is that overall survival (OS) is prolonged in participants who receive pembrolizumab compared to those receiving placebo. * **Secondary Objectives:** To evaluate Progression-Free Survival (PFS), Objective Response Rate (ORR), and Duration of Response (DOR) based on RECIST version 1.1.

2.2 Background & Rationale Hepatocellular carcinoma (HCC) is the most common form of primary liver cancer and a leading cause of cancer-related mortality globally. Patients with advanced HCC who have progressed on or are intolerant to first-line therapies (such as sorafenib or oxaliplatin-based chemotherapy) have historically faced poor prognoses and limited treatment options. This study aims to address the unmet medical need by confirming the clinical benefit of pembrolizumab, an anti-PD-1 immunotherapy, in the second-line setting for Asian patients.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Placebo-controlled (Placebo + Best Supportive Care) * **Blinding:** Double-blind * **Randomization:** 2:1 ratio (pembrolizumab : placebo)

3.2 Planned Enrollment & Duration * **Targeted Enrollment:** 453 participants enrolled and randomized * **Number of Sites:** Multicenter study in Asian countries * **Trial Status:** COMPLETED

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Has a HCC diagnosis confirmed by radiology, histology, or cytology (fibrolamellar, and mixed hepatocellular/cholangiocarcinoma subtypes are not eligible). 2. Has Barcelona Clinic Liver Cancer (BCLC) Stage C disease or BCLC Stage B disease not amenable to locoregional therapy or refractory to locoregional therapy and not amenable to a curative treatment approach. 3. Has a Child-Pugh A liver score within 7 days prior to first dose of study medication. 4. Has a life expectancy of >3 months. 5. Has at least one measurable lesion based on RECIST version 1.1 as determined by investigator. 6. Has Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1 performed within 7 days prior to receiving the first dose of study medication. 7. Has documented objective radiographic progression during or after treatment with sorafenib or oxaliplatin-based chemotherapy, or else intolerance to sorafenib or oxaliplatin-based chemotherapy. 8. Female participants of childbearing potential must have a negative urine or serum pregnancy test within 72 hours prior to receiving the first dose of study therapy. 9. Female and male participants of reproductive potential must agree to use adequate contraception starting from the first dose of study medication, throughout the study period, and for up to 120 days after the last dose of study medication.

4.2 Exclusion Criteria 1. Is currently participating or has participated in a study with an investigational agent or using an investigational device within 4 weeks of the first dose of study medication. 2. Has received sorafenib or oxaliplatin-based chemotherapy within 14 days of first dose of study medication. 3. Has had esophageal or gastric variceal bleeding within the last 6 months. 4. Has clinically apparent ascites on physical examination. 5. Has portal vein invasion at the main portal branch (Vp4), inferior vena cava, or cardiac involvement of HCC based on imaging. 6. Has had clinically diagnosed hepatic encephalopathy in the last 6 months. 7. Has had a solid organ or hematologic transplant. 8. Has had prior systemic therapy for HCC in the advanced (incurable) setting other than sorafenib or oxaliplatin-based chemotherapy, prior to start of study medication. 9. Has an active autoimmune disease that has required systemic treatment in the past 2 years. Replacement therapy is not considered a form of systemic treatment. 10. Has a diagnosis of immunodeficiency or is receiving systemic steroid therapy or any other form of immunosuppressive therapy within 7 days prior to the first dose of study medication. 11. Has received locoregional therapy to liver (transcatheter chemoembolization [TACE], transcatheter embolization [TAE], hepatic arterial infusion [HAI], radiation, radioembolization, or ablation) within

4 weeks prior to the first dose of study medication. 12. Has had major surgery to liver or other site within 4 weeks prior to the first dose of study medication. 13. Has had a minor surgery ≤ 7 days prior to the first dose of study medication. 14. Has not recovered adequately (i.e., Grade ≤ 1 or baseline) from the toxicity and/or complications from any intervention prior to study start. 15. Has a diagnosed additional malignancy within 3 years prior to first dose of study medication with the exception of curatively treated basal cell carcinoma of the skin, squamous cell carcinoma of the skin and/or curatively resected in situ cancers. 16. Has a known history of, or any evidence of, central nervous system (CNS) metastases and/or carcinomatous meningitis. 17. Has a history of (non-infectious) pneumonitis that required steroids or current pneumonitis. 18. Has an active infection requiring systemic therapy. 19. Is pregnant or breast feeding or expecting to conceive or father starting from the first dose of study medication, throughout the study period, and for up to 120 days after the last dose of study medication. 20. Has received prior immunotherapy with an anti-Programmed Cell Death Receptor 1 (PD-1), Programmed Cell Death Receptor Ligand 1 (anti-PD-L1), or anti-Programmed Cell Death Receptor Ligand 2 (PD-L2) or has previously participated in clinical studies with pembrolizumab. 21. Has a known history of human immunodeficiency virus (HIV). 22. Has untreated active Hepatitis B. 23. Has Hepatitis C in which participants received therapy for HCV < 4 weeks prior to receiving pembrolizumab. 24. Has received a live vaccine within 30 days prior to the first dose of study therapy.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * Dosage/Frequency: Pembrolizumab 200 mg or saline placebo administered once every 3 weeks. * **Route of Administration:** Intravenous (IV) infusion. * **Duration of Treatment:** Up to 35 cycles (approximately 2 years) or until documented disease progression, unacceptable toxicity, or withdrawal.

5.2 Concomitant & Prohibited Therapy * Permitted: Best Supportive Care (BSC) as per local guidelines, including pain management and management of other potential complications like ascites. * **Prohibited:** Other investigational agents, device trials, or unapproved concurrent systemic anti-cancer therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to Day -1	Informed Consent, Eligibility screening, baseline tumor imaging, Safety Labs
Baseline	Day 1, Cycle 1	Randomization, First Dose (Pembrolizumab/Placebo + BSC)
Interim	Every 3 Weeks	IV Infusion, Safety Labs, Efficacy/Imaging Assessments (Tumor

imaging every 6 weeks for year 1, then every 9 weeks) | | **End of Study** | Up to 35 Cycles | End of Treatment Visit, Survival Follow-Up phase |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * Overall Survival (OS): OS is the time from randomization to death due to any cause, based on the Kaplan-Meier method for censored data.

7.2 Secondary Endpoints * Progression Free Survival (PFS): Per Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST 1.1). * **Objective Response Rate (ORR):** Per Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST 1.1). * **Duration Of Response (DOR):** Per Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST 1.1).

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Continuous AE monitoring throughout the treatment period and specific safety follow-up. Special focus on immune-mediated adverse events.
 - **Serious Adverse Events (SAEs):** Standard reporting timelines as per GCP, measuring the number of participants experiencing at least one AE/SAE.
 - **Data Monitoring Committee (DMC):** An independent external data monitoring committee assessed safety and efficacy continuously and at planned interim analyses.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) population for primary efficacy outcomes (OS, PFS).
 - **Sample Size Justification:** $N=453$ appropriately powered (one-sided significance thresholds set at $P = 0.0193$ for OS and $P = 0.0134$ for PFS) to detect meaningful survival benefits.
 - **Handling of Missing Data:** Censoring rules applied via standard Kaplan-Meier methodology for time-to-event endpoints.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study was conducted in accordance with Good Clinical Practice (GCP) guidelines and the Declaration of Helsinki.

- **Monitoring Plan:** Strict oversight by an independent external data monitoring committee, with site monitoring conducted by the Sponsor.
 - **Audit Trail:** Robust centralized data collection, utilizing Blinded Independent Central Review (BICR) for all crucial radiological progression assessments.
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11. Study Identifiers & Governance

Primary ID: MK-3475-394 **Registry Number:** NCT03062358 **Sponsor/Lead Organization:** Merck **Study Title:** KEYNOTE-394 **Official Regulatory Title:** A Phase III Randomized Double-blind Study of Pembrolizumab plus Best Supportive Care vs. Placebo plus Best Supportive Care as Second-Line Therapy in Asian Subjects with Previously Systemically Treated Advanced Hepatocellular Carcinoma

12. Clinical Framework & Terminology

Primary Condition: Carcinoma, Hepatocellular **Preferred UMLS Name:** Carcinoma, Hepatocellular **Semantic Type:** Neoplastic Process
Disease Definition: A malignant tumor arising from epithelial cells. Carcinomas that arise from glandular epithelium are called adenocarcinomas, those that arise from squamous epithelium are called squamous cell carcinomas, and those that arise from transitional epithelium are called transitional cell carcinomas.
Medical Methodology: Anti-PD-1 Immunotherapy + Best Supportive Care **Ontology Codes:** * **MeSH Code:** D002277 * **HPO Code:** HP:0030731 * **SNOMED ID:** 1187425009 (Hierarchy: 1187225007 | Malignant epithelial neoplasm, 118285006 | Epithelial neoplasm, 108369006 | Neoplasm, 1240414004 | Malignant neoplasm)

13. Study Procedures & Methodology

Management Pathway: Systemic second-line immuno-oncology therapy
Education Protocol (HCP): Clinical trial site initiation training; protocol compliance for RECIST v1.1 reporting
Education Protocol (Patient): Informed consent, immune-mediated AE awareness
Quality Audit Mechanism: Blinded Independent Central Review (BICR) of radiographic evidence

14. Observation & Follow-Up Schedule

Total Duration: Up to ~2 years (35 treatment cycles) + continuous survival follow-up
Onsite Visit Cadence: Every 3 weeks for treatment dosing
Remote Monitoring Frequency: Intermittent survival check-ins post-treatment
Checklist Review Interval: Tumor imaging every 6 weeks for the first year; every 9 weeks thereafter

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]					